



Harnessing non-noble metal-doped carbon composites for remediation of phthalate esters contaminated water systems

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ABSTRACT

Conventional noble metal catalysts exhibit excellent electrical conductivity and catalytic activity; however, their high cost, moderate cyclability, and limited availability render them economically impractical for widespread applications. To solve these problems, non-noble metal-doped carbon (Met@C) composites are highly active, economical, structurally and chemically stable, and have been recommended as next-generation materials for removal of emerging contaminants, such as phthalate esters (PAEs). PAEs are commonly used as plasticizers and categorized as low- and high-molecular weight PAEs to understand their differential acute and chronic toxicological profiles. However, they may leach from plastic products, raising environmental concerns regarding their use. A review of PAEs-Met@C interactions will be helpful for targeted design of Met@C for effective PAEs removal. This paper reviews Met@C and its synthesis and environmental applications, specifically PAEs removal from water via adsorption, activation oxidation, photocatalysis, Fenton/Fenton-like processes, and coupled treatment processes and the mechanisms overlaid. It also discusses factors influencing the performance of Met@C for PAEs removal and degradation. To provide mechanistic insights into degradation kinetics and mechanisms of AOPs for PAEs removal, it describes pseudo-1st-order and 2nd-order rate constants and radical and nonradical species along with density functional theory and molecular dynamics simulations. To address stability, durability, and scalability of Met@C, it thoroughly explores metal leaching, regeneration, and real-world applications of Met@C for PAEs removal. Before conclusions, it overviews perspectives, future challenges, and opportunities for PAEs removal by Met@C. Overall, this review provides a basis for Met@C to solve technique problems in remediation of PAEs contaminated water systems.

1. Introduction

1.1. Non-noble metal-doped carbon composites

Carbon (C)-based materials are pivotal next-generation materials due to their exceptional physicochemical properties, including high thermal, mechanical, and electrical conductivities, which enable them

to drive advancements across various fields, such as biosensing, composite reinforcement, conductive materials, drug delivery, biomedicine, catalysis, and catalyst support [1–3]. Additionally, C-based materials are highly effective for inorganic and organic pollutant removal [4,5]. Although noble metals such as Pt, Pd, Ru, Rh, Re, and Ir demonstrate exceptional electrical conductivity and catalytic activity [6,7], their high cost, moderate cyclability, and limited availability render them

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economically impractical for widespread applications. Hence, it is crucial to identify cost-effective alternatives that offer comparable or superior efficiency in terms of cyclability and stability [8]. Composites containing earth-abundant non-noble metals (Fe, Ni, Co, Cu, Al, Ti, Mn, and Zn) are promising alternatives to catalysts containing noble metals [9,10]. Non-noble metal cations and oxides serve as active components in environmental catalysis and are considered benchmark catalysts for advanced oxidation processes (AOPs) in both homogeneous and heterogeneous systems [11–13]. Their applicability has been demonstrated with structure-dependent catalysis of Cu_2O [14], Co^{2+} -mediated [11], Co_3O_4 for heterogeneous activation [15], amorphous MnO_2 [16], oxygen vacancy-laden NiO [12], and Fe_3O_4 magnetic nanoparticles [13], showing enhanced catalytic removal of organic compounds (OCs). Recently, mono-, di-, and multi-non-noble metal alloys have been extensively studied as alternatives to noble metal catalysts for remediation of water systems. Engineered alloy compositions and dual-metal-doped C-based catalysts, rather than single-metal catalysts, exhibit improved catalytic performance owing to grain boundary strain, particular functionalities, and synergistic effects [17].

It is widely accepted that C-based materials have stable structures and better conductivity but low removal efficiency, and non-noble metal-based materials have high activity but are prone to aggregation and efficiency reduction. Combining C-based materials with non-noble metals can provide hybrid composites that integrate high activity of metals with exceptional conductivity and stability of C, and exhibit new physicochemical characteristics owing to synergy of doped materials and persistence of doped structures, thereby providing highly active composites [18]. Non-noble metal-doped carbon (Met@C) composites are highly active, economical, and structurally and chemically stable, and C-support enhances metal dispersion, prevents aggregation and leaching, and improves electron (e^-) transfer efficiency and mass transport because of their porous structure and strong Met-C interactions [19,20]. Met@C is capable of maintaining separation of e^- -hole (h^+) pairs, e^- transfer efficiency, and free radical generation to degrade OCs through AOPs [21–23]. Kang et al. studied physicochemical properties and efficacy of Met@C hybrids comprising activated carbon (AC) and metal oxides (Fe_3O_4 and MnO_2) to explain adsorption and degradation mechanisms of polycyclic aromatic hydrocarbons (PAHs) in water [24]. Furthermore, nanosized metals supported on ceria-granular C have been utilized to remove OCs from air [25] and C-metal oxide hybrids to remove OCs from water [24]. Met@C has an excess of active sites with better functionalities, high surface area, low metal leaching, locally unpaired e^- , strong Met-C-support interactions, and better performance for OCs removal [26].

AOPs have been widely studied for removing persistent OCs from water using different oxidants namely hydrogen peroxide (H_2O_2) and persulfate (PS) which mainly include peroxymonosulfate (PMS, HSO_5^-) and peroxydisulfate (PDS, $\text{S}_2\text{O}_8^{2-}$) [10]. C-based materials and MWCNTs doped with non-noble metal oxides/hydroxides/sulfides have emerged as activators for removing PAEs in AOPs [21,27–29]. Nanoscale CuFe_2O_4 /MWCNTs effectively activated PDS to produce sulfate radicals ($\text{SO}_4^{\cdot -}$) for phthalate esters (PAEs) degradation [27]. Commercial MWCNTs, synthesized via chemical vapor deposition, often contain carbonaceous by-products and impurities and metal catalyst particles (e.g., Al, Ni, Co, and Mn), which reduce adsorption efficiency and require their removal. Poor dispersion and bundling of CNTs due to van der Waals interactions further limit their applications [30,31]. To address these issues, naturally abundant agricultural feedstocks can be used as C-supports and sources for Met@C synthesis. Carbon nanofiber (CNF) derived from cellulose was used to support CoFe_2O_4 to activate PMS, demonstrating excellent catalytic performance of CoFe_2O_4 @CNF for PAEs removal [32]. Met@C in which C derived from abundant agricultural feedstocks has demonstrated effective remediation of PAEs contaminated water systems [33–35].

Despite their advantages in terms of efficiency and reactivity, Met@C faces challenges such as complex architecture and non-uniform

dispersion, migration, and aggregation of metals [17]. For example, metal doping may occupy active adsorption sites or block pores, resulting in low surface area, short-term activity, aggregation of metals, and low removal performance of Met@C [26]. Suitable approaches for overcoming these problems include optimizing synthesis and functionalizing Met@C to enhance catalytic efficiency by adjusting metal ions dispersion on C-support. Synthesis of Met@C requires advanced but straightforward processes to ensure uniformity and explicit coordination between non-noble metals, dopants, and C-supports. Current characterization techniques limit in-depth understanding of structure-property relationships. Theoretical modeling plays a crucial role in complementing experimental studies, helping rationalize observations, selecting optimal metal and C combinations, and reducing research time and costs [36]. Theoretical computational calculations, such as density functional theory (DFT) and molecular dynamics (MD) simulations, make production of Met@C more effective by providing atomic-scale insights into active sites and formation mechanisms, enabling prediction and optimization of synthesis parameters, and bridging gap between Met@C design and practical performance (activity, selectivity, and stability) for catalysis, energy storage, and sensing. Together, DFT and MD simulations transform Met@C development from empirical to predictive, accelerating deployment of low-cost and high-performance alternatives for PAEs removal and elucidating mechanisms underlying PAEs removal by Met@C [36–38]. With optimized theoretical and unconventional experimental parameters, Met@C can effectively remove OCs via adsorption and AOPs [21,39,40].

Functionalization of Met@C is a highly effective strategy that greatly impacts Met-C interactions by precisely adjusting electronic and chemical characteristics and surface structures of C-supports, eventually enhancing catalytic efficiency and selectivity of systems [41,42]. Furthermore, local coordination structures of functionalities, dispersion of different metal species and their stability, and evolution of active sites from raw precursors to final C-based materials during carbonization play crucial roles in environmental applications [43,44]. Functionalized Met@C, such as macroporous $\text{H}_5\text{PMo}_{10}\text{V}_2\text{O}_{40}(44)/\text{BC}$ [45], magnetic Fe oxide/GO ($\text{Fe}^{\text{III}}\text{Fe}_2^{\text{II}}\text{O}_4/\text{GO}$) [39], 3D N-Co@C/hierarchical C frameworks [46], graphene quantum dots/Mn-N-TiO₂/g-C₃N₄ [21], and Nb₂C MXene/ Fe^{3+} [47], has demonstrated satisfactory remediation of PAEs contaminated water systems.

1.2. PAEs and their removal challenges

PAEs, a type of hydrophobic OCs, have attracted public attention because of their widespread use as plasticizers, which are additives in plastics (mainly polyvinyl chloride (PVC)) that increase their durability and flexibility [48,49]. PAEs are colorless sticky liquids characterized by hydrocarbon functional groups ranging from 1 to 13 carbons. Approximately 60 distinct PAEs and their by-products are consumed as additives in cosmetics, adhesives, lubricants, packaging, insecticides, and paint industries [50]. The dimethyl phthalate (DMP), diethyl phthalate (DEP), diallyl phthalate (DAP), dibutyl phthalate (DBP), di-n-butyl phthalate (DnBP), diisobutyl phthalate (DIBP), benzyl butyl phthalate (BBP), diisohexyl phthalate (DIHP), di(2-ethylhexyl) phthalate (DEHP), di-n-octyl phthalate (DnOP), diisononyl phthalate (DINP), and diisodecyl phthalate (DIDP) are commonly used and toxic PAEs [22]. They are categorized as low- and high-molecular weight PAEs to understand their differential acute and chronic toxicological profiles and removal processes. Low-molecular weight (LMW, short alkyl chains (1–4 C atoms) with molecular weight of <300 g/mol; DMP, DEP, DnBP, and DIBP) PAEs are more volatile and water-soluble and have applications in personal care products, cosmetics, pharmaceuticals, insecticides, perfumes, adhesives, waxes, nail polishes, and printing inks [51,52]. High-molecular weight (HMW, long alkyl chains (≥ 5 C atoms) with molecular weight of >300 g/mol; DEHP, DINP, DIDP, and BBP) PAEs are more lipophilic and less water-soluble. Approximately 80% of alkyl groups with long-chain PAEs have applications in PVC industry,

flooring, wall coverings, packaging, and medical devices [51,53].

Worldwide production of PAEs rose from 2.7 to 6 Mt/annum from 2007 to 2017 [54]. However, extensive use of PAEs has increased their occurrence in environment, as they are physically bonded with plastic matrices through weak interactions and easily leach from plastics when exposed to water or organics and subjected to surface contact and mechanical pressure [55]. Free PAEs can enter humans through water, food chain, and skin contact, causing destructive health effects [56]. PAEs are present in air, water, and soil at detectable concentrations, and 23 types of PAEs are recognized as dangerous substances by World Health Organization, and U.S. EPA recommends monitoring PAEs in drinking water at concentrations >0.6 mg/L [49,57,58]. Fig. 1 demonstrates the timeline of key developments in the history of PAEs from 1920s to 2020s, highlighting evolution from commercial production to environmental persistence to modern sustainability efforts.

Due to mutagenic, carcinogenic, and endocrine-disrupting properties of PAEs, substantial steps have been taken to develop suitable removal materials and processes to understand their removal and degradation mechanisms. Main goal of PAEs remediation systems is to realize high performance (of Met@C) while maintaining low metal contamination, minimizing chemical input, and reducing C emissions [33,59]. Adsorption, AOPs, such as activation oxidation, photocatalysis, Fenton/Fenton-like processes, and coupled treatment processes have shown satisfactory remediation of PAEs contaminated water systems [23,47,56,60]. However, each removal process has its own challenges and limitations that must be addressed. For instance, release of OCs once adsorption reaches saturation, disposal of used adsorbate, and longer operation time are inadequacies for adsorption of PAEs [61]. Activation oxidation is a chemically stable process that effectively degrades PAEs at flexible pH (2–11), uncertainties remain regarding toxicity of intermediate compounds, formation of unwanted halogenated by-products, and role of primary and secondary radicals [26]. Despite being one of the most favorable AOPs for degrading OCs, low sunlight usage, migration ability, and chemical stability are challenges for photocatalysis of PAEs [62]. Classic Fenton reaction ($\text{Fe}^{2+}/\text{H}_2\text{O}_2$) generates hydroxyl radicals ($\cdot\text{OH}$) with a short lifetime of <1 μs in harsh acidic environments. Fenton/Fenton-like processes for PAEs removal are hindered by instability of H_2O_2 /ferrous salt, leading to challenges such as Fe^{2+} loss and

metal contamination, which necessitate post-treatment to manage and recycle sludge [63]. This review provides insights into challenges of PAEs removal and reactivity trends of Met@C, offering a basis for optimized removal strategies and future coupled treatment process designs.

1.3. Purpose of this review

A search using keyword “carbon materials for phthalate esters” in Web of Science Core Collection yielded over 150 articles, including 141 research papers and 11 review articles. In 2008, a pioneering study initiated research on removal of PAEs by C-based materials, where bamboo charcoal was employed as an adsorbent to detect 4 PAEs in environmental waters [64]. Following 2010s, research on PAEs removal by C-based materials grew steadily, with publications from 2014 to 2024 on this topic are illustrated in Fig. S1. There are reviews on the occurrence [57,65], toxicity [66], environmental fate [67], and removal processes [68,69] of PAEs. Moreover, electrochemical methods utilizing functional nanomaterials [70] and AOPs utilizing BC and transition metal systems [68] have been reviewed for detection and removal of PAEs. A diverse array of PAEs underscores need for robust water treatment processes, which can be achieved by next-generation materials such as Met@C. Earlier, Met@C has been reviewed for heterogeneous catalysis [71], electrocatalysis for water splitting [72], water electrolysis [73], and energy applications [74]. However, a specific review on Met@C for PAEs removal is lacking to date. Thus, this paper reviews the role of Met@C for in-depth exploration of PAEs removal and the underlying mechanisms. Inherent challenges of PAEs removal by Met@C include nuanced intricacies of mechanisms and misunderstandings of removal reactions, which are the focal points of this review. Objectives of this review are to (i) overview PAEs and their removal challenges; (ii) discuss adsorption, AOPs, and coupled treatment processes for PAEs removal and the mechanisms overlaid; (iii) explore degradation kinetics and mechanisms of AOPs for PAEs removal via theoretical modeling and experimental studies; and (iv) understand metal ions leaching, regeneration, real-world applications, and cost evaluation of Met@C for PAEs removal.

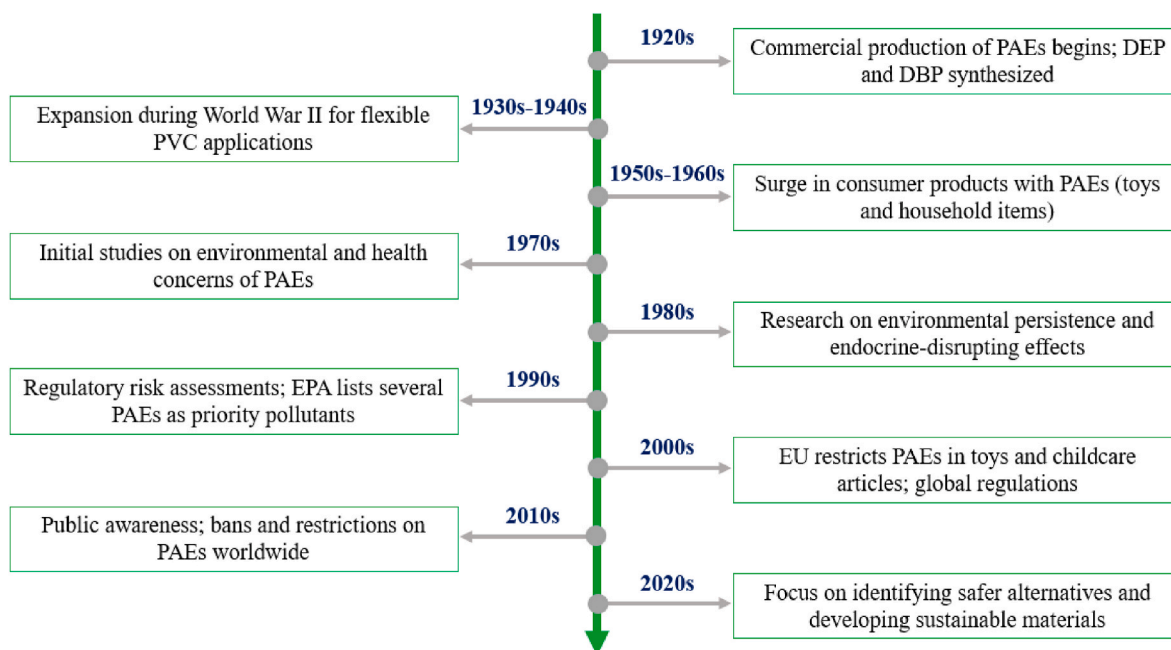


Fig. 1. Timeline illustrates significant milestones in the history of PAEs, from their initial commercial production in 1920s to contemporary advancements in sustainable alternatives by 2020s.

2. Synthesis of Met@C

Choice of synthesis approach greatly affects conductivity, acidic/basic characteristics, oxidation stability, and performance of Met@C in environmental and energy applications [76]. Common approaches for synthesizing Met@C include pre-modification of C-based materials, *in-situ* doping (direct carbonization), and post-modification of as-synthesized C-based materials [42,77]. Each synthesis approach encounters challenges that must be addressed to achieve optimal synthesis of Met@C (Fig. 2) [42,78,79]. Detailed synthesis of Met@C is described in Text S1 of Supplementary Information.

3. Met@C for remediation of PAEs contaminated water systems

Classifying PAEs by their removal mechanisms – adsorption, activation oxidation, photocatalysis, and Fenton/Fenton-like processes – creates a remediation-focused framework linking physicochemical traits of PAEs to treatment reactivity of Met@C (Tables 1 and 3). This classification aids in process selection, reveals compound-specific limitations, and informs optimized strategies and coupled treatment processes for environmental remediation of PAEs.

3.1. Adsorption

Application of Met@C has demonstrated promising efficacy in adsorption of PAEs due to its improved efficiency, affordability, reusability, and ease of operation [31,56]. Fig. 3a shows removal mechanisms of PAEs by Met@C via adsorption. Met@C [80–82] demonstrated superior performance in adsorption of PAEs compared to individual C-based materials such as commercially available AC [61,83], BC and nanoscale BC [84–86], graphene [87], graphene oxide (GO) [88], CNTs [89], and chitosan [90], as well as non-noble metal cations and oxides [91,92] (Fig. 4a, Table S1). Fig. 4a indicates that Fe is the most widely used non-noble metal to synthesize Met@C for adsorption of PAEs [82, 93,94]. Superior performance of Met@C is attributed to the fact that particles of non-noble metals on C layer/support experience less dissolution of metal ions and oxidation of Met@C, and Met-C interactions provide better active sites to efficiently remove PAEs. PAEs inspected for adsorption by Met@C included individual PAEs such as DMP [31,82],

DEP [39,61,80], and DnBP [35], as well as multiple PAEs [46,81,95] (Table 1). It is observed that moderate removal of LMW PAEs was due to their higher water solubility, which reduced adsorption, whereas functionalization of Met@C enhanced PAEs removal [31,80]. High removal of HMW PAEs indicates that strong hydrophobicity enhanced adsorption on C-based materials [56,81].

Adsorption isotherms are crucial for understanding and predicting system performance under varying conditions, optimizing process design, and handling fluctuating PAEs concentrations [31,80]. To examine adsorption process for PAEs removal by Met@C, two-parameter adsorption isotherms, namely Langmuir [61,80,82] and Freundlich [31,35] models are most promising ones (Table 1). Adsorption kinetics are associated with both PAEs properties and Met@C chemistry/structures. For adsorption kinetics, pseudo-2nd-order model [31,35,61,95] is most effective for PAEs removal. Adsorption kinetics and capacities are both crucial for efficient PAEs removal, and their importance depends on application [35,81,95]. Adsorption kinetics, which describe how quickly PAEs are removed [31], are vital for rapid treatment needs, such as industrial wastewater or emergency response. Adsorption capacity, maximum PAEs adsorbed per unit mass, is key for maximizing removal efficiency and minimizing adsorbent replacement [81], prioritized in slower and high-efficiency processes.

PAEs have diverse physicochemical properties influenced by carbon chain structures (CCS) of their alcohol components, affecting their hydrophobicity, molecular volume, and π -electron density [56,96]. Longer or branched carbon chains increase molecular volume and hydrophobicity but reduce π -electron density, thereby altering removal mechanisms (Fig. 3a). For instance, among DMP, DEP, DnBP, DnOP, and DEHP, DnOP exhibited fastest adsorption, while DEHP was slowest due to its branched structure, higher molecular volume, and increased water solubility compared to DnOP [56]. Although both DnOP and DEHP contain 8 C atoms, difference in their removal was attributed to their CCS. With identical C number of CCS, branched chains promote PAEs aggregation, reducing adsorption efficiency [56]. Microscopic studies showed that DMP, DEP, and DnBP were primarily removed via π - π electron donor-acceptor (EDA) interactions, DEHP through hydrophobic interactions, and DnOP through a combination of both (Fig. S2a).

For chemisorption of PAEs on C-based materials, for short CCS PAEs, key removal mechanisms of PAEs on porous C-based materials are π - π

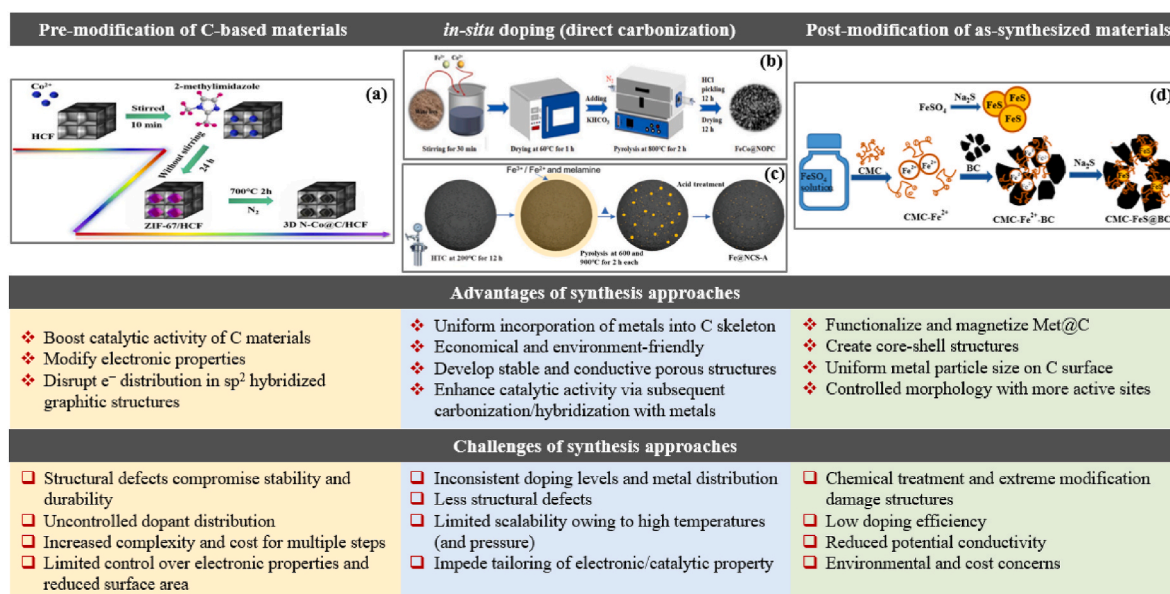


Fig. 2. Synthesis approaches for Met@C, including typical examples, along with their advantages and challenges; pre-modification of C-based materials: 3D N-Co@C/HCF (a) [46], *in-situ* doping (direct carbonization): Fe/Co@NOPC (b) [40] and Fe@NCS-A (c) [44], and post-modification of as-synthesized C-based materials: CMC-FeS@BC (d) [75].

Table 1
Precursors, operation conditions, and removal performance of PAEs via adsorption by Met@C.

Met@C	Precursor	PAEs	Operation conditions	Removal capacity/efficiency	Isotherms/kinetics/key findings	Ref.
Fe@C	D201 resin, K ₂ C ₂ O ₄ , Fe(NO ₃) ₃	DEP	[DEP] ₀ : 12 mg/L, dose: 1 mL, pH: 4–11, T: 25 °C	714.29 mg/g removal capacity	- Langmuir and pseudo-2nd-order models - Spontaneous and exothermic adsorption	[61]
MWCNTs@Fe ₃ O ₄ @SiO ₂	MWCNTs, Fe ₃ O ₄ , SiO ₂ , NH ₃ ·H ₂ O	DMP, DEP, DnBP	[PAEs] ₀ : 5 mg/L, dose: 10 g/L, pH: 6–7, T: 25 °C	0.95, 1.38, and 7.09 mg/g removal capacities for DMP, DEP, and DnBP, resp.	- Langmuir and pseudo-2nd-order models - Spontaneous and endothermic adsorption - Physisorption based on H-bonding, hydrophobic interaction, and van der Waals force	[95]
Fe ₃ O ₄ /MWCNTs	FeSO ₄ ·7H ₂ O, Fe ₂ (SO ₄) ₃ , MWCNTs	DMP	[DMP] ₀ : 40 mg/L, dose: 0.2 g/L, pH: 7, T: 25 °C	136.99 mg/g removal capacity	- Freundlich and pseudo-2nd-order models - Spontaneous and endothermic adsorption	[31]
BC@MnO ₂	Corn stalk, MnSO ₄ ·H ₂ O, KMnO ₄	DnBP	[DnBP] ₀ : 0.0215 mmol/L, dose: 0.02 g, T: 25 °C	3.639×10 ⁻² mmol/g removal capacity	Langmuir, Freundlich, and pseudo-2nd-order models	[35]
Activated Fe@C aerogel	Resorcinol, Na ₂ CO ₃ , ferric acetylacetonate	DMP	[DMP] ₀ : 50 mg/L, pH: 6	84% removal efficiency	- π–π EDA interactions - Electrostatic interaction, hydrophobic interaction, π–π EDA interactions, and molecular imprinting affinity	[82]
Fe ₃ O ₄ @voidC	Fe ₃ O ₄ , resorcinol, NH ₃ ·H ₂ O	DEP, DnBP, BBP, DEHP, DnOP	[PAEs] ₀ : 100–1000 mg/L, dose: 10 mg, T: 25 °C	117.5, 96.0, 112.1, 119.8, and 123.9 mg/g removal capacities for DEP, DnBP, BBP, DEHP, and DnOP, resp.	Uniformity and monodispersity of Fe₃O₄@voidC showed better adsorption of PAEs	[81]
GO-NH ₂ -MNP	FeCl ₃ ·6H ₂ O, CH ₃ COONa, 1,6-hexanediamine	DEP	[DEP] ₀ : 50 mg/L, dose: 2 g/L, pH: 3–10	8.71 mg/g removal capacity	Langmuir and pseudo-2nd-order models	[80]
3D N-Co@C/HCF	Glucose, sodium carbonate, Co(NO ₃) ₂ ·6H ₂ O	DMP, DEP, DnBP, BBP, DCHP	[PAEs] ₀ : 1 mg/L, dose: 30 mg	87.1–107.2% recovery of PAEs	- π–π EDA interactions - Satisfactory recoveries of PAEs in real samples	[46]
Fe ^{II} Fe ^{III} O ₄ @GO	FeCl ₃ ·6H ₂ O, FeCl ₂ ·4H ₂ O, GO, NaOH	DEP	[DEP] ₀ : 4.21 mg/L, dose: 334.7 mg/L, sonication time: 3.723 min, pH: 5.38	116.93 mg/g removal capacity	- Langmuir and pseudo-2nd-order models - Spontaneous and endothermic adsorption	[39]

Note: []₀: Initial concentration, DCHP: Dicyclohexyl phthalate.

Table 2
Comparative overview of oxidation, photocatalysis, and Fenton/Fenton-like processes.

Characteristics	Activation oxidation	Photocatalysis	Fenton processes
Catalyst stability	Chemically stable (with C support) [22,97]	High stability (e.g., TiO ₂) [21,30]	Limited stability; Fe cycling, unstable H ₂ O ₂ /Fe ²⁺
Radicals	Mainly SO ₄ ^{•-} , some •OH [98]	Mainly •OH [99]	Mainly •OH [100]
Metal contamination	Low metal leaching	Contamination from semiconductor metals	Significant due to Fe ²⁺ /Fe ³⁺
Selectivity	High oxidative potential, selective oxidation	Moderate selectivity	Low selectivity (unstable H ₂ O ₂)
Chemical input	Met@C activators	UV light and Met@C	H ₂ O ₂ and Fe ²⁺
pH	Wide pH range (2–11)	pH-dependent efficiency	Harsh acidic conditions (pH~3)
Toxicity	Generally low-toxic by-products	Minimal toxic by-products	Possible Fe sludge and intermediates
Reaction mechanism	Activation of PS/H ₂ O ₂ to ROS	Light-induced e ⁻ excitation in Met@C	Fenton reaction: Fe ²⁺ /H ₂ O ₂
Target PAEs	LMW PAEs [27,32] to HMW PAEs [34,101]	DMP [28,102], DEP [21,29], DnBP [103,104]	LMW PAEs (e.g., DMP, DEP) [40,105]
Removal performance	High removal of LMW PAEs; faster reaction kinetics, moderate removal of HMW PAEs; limited reaction kinetics [34]	Moderate-high removal for LMW PAEs; high solubility and diffusion on Met@C, Less availability of trusted data for HMW PAEs [21,30]	Moderate-high removal for LMW PAEs; breaks O–O bond and charge (e ⁻) transfer from Met@C to oxidant [37,106]
Cost	Medium-high; depends on activation method	High; light source, Met@C cost, maintenance	Low-medium; cheap chemicals, increase with post-treatment

EDA interactions and pore filling. Considering graphitization degree of C in N-Co@C/hierarchical C framework, N-doping benefits formation of π-conjugated effect due to activation of π-electrons of sp² C in N-Co@C/hierarchical C framework by lone pair e⁻ of N atoms. Thus, π–π conjugation could be a key mechanistic interaction between N-Co@C/hierarchical C framework and PAEs [46]. With increasing CCS length, π–π EDA interactions and pore filling could decline; in contrast, hydrophobic interactions could increase. Hydrophobic surfaces of Met@C offer uniformly distributed hydrophobic sites for PAEs [46]. Hydrophobic parameter (log K_{OW}) of OCs can serve as indicator to gauge hydrophobic

interaction capabilities. For instance, log K_{OW} of DMP was 1.53, indicating that hydrophobic interactions could be a key mechanism for DMP adsorption on activated Fe@C aerogel [82]. PAEs resist water solubility due to their hydrophobic nature, and higher wettability of Met@C improves interactions with aqueous solutions, enhancing contact and efficient removal of hydrophobic PAEs [31,39,86]. However, balancing wettability of Met@C and hydrophobic interactions with PAEs is crucial for optimal performance.

Table 3

Precursors, operation conditions, and removal performance of PAEs via AOPs by Met@C.

Removal method	Met@C	Precursor	PAEs	Operation conditions	TOC mineralization	Degradation rate and key findings	Ref.
Activation oxidation	CuFe ₂ O ₄ /MWCNTs	Cu(NO ₃) ₂ ·3H ₂ O, Fe(NO ₃) ₃ ·9H ₂ O, MWCNTs-COOH	DEP	[DEP] ₀ : 5 mg/L, dose: 0.1 g/L, DEP/PDS: 1/100, pH: 7.0 ± 0.2, T: 25 ± 2 °C	23% in 0.5 h	Complete degradation in 0.5 h; SO ₄ ^{•−} —was major radical	[27]
	G-CoFe ₂ O ₄	GO, Fe(NO ₃) ₃ ·9H ₂ O, Co(NO ₃) ₂ ·6H ₂ O	DMP	[DMP] ₀ : 0.05 mM, [PMS] ₀ : 2 mM, dose: 0.5 g/L, pH: 4–8.35, T: 23 °C	≈24% in 1 h	Complete degradation in 0.5 h; SO ₄ ^{•−} and •OH attack of aromatic ring and oxidation of aliphatic chain	[98]
	CoFe ₂ O ₄ @CNF	Cellulose acetate, Fe(NO ₃) ₃ ·9H ₂ O, Co(NO ₃) ₂ ·6H ₂ O	DMP	[DMP] ₀ : 0.05 mM, [PMS] ₀ : 1.5 mM, dose: 0.5 g/L, pH: 3.18–9.07	75% in 2 h	Complete degradation in 0.5 h; SO ₄ ^{•−} was main ROS	[32]
	H ₅ PMo ₁₀ V ₂ O ₄₀ @Surf/GO	GO, HPMoV	DEP, DAP, DEHP	[PAE] ₀ : 100 mg/L, [H ₂ O ₂] ₀ : 1.3 g/L, dose: 6.0 g/L, pH: 7, T: 25 °C	>70%	93% degradation; •OH, O ₂ ^{•−} , and ¹ O ₂ were main ROS	[116]
	Fe-Ce@BC	Fe(NO ₃) ₃ , Ce(NO ₃) ₃ , water caltrop shell biomass	DMP, DEP, DIBP, DnBP, DIHP, BBP, DNP, DEHP, DINP, DIDP	Sediment: 1 g, dose: 1.7 g/L, pH: 6, [PDS] ₀ : 1 × 10 ^{−5} M, PDS/PAE: 1/1, T: 30 °C	–	Electrostatic and π-π EDA interactions with PDS activated by Fe ³⁺ /Fe ²⁺ and Ce ⁴⁺ /Ce ³⁺ redox cycles on surface were main degradation mechanisms	[22]
	Fe ₃ O ₄ @BC	Rice husk, Fe ₃ O ₄ , NH ₃ , HNO ₃	DnBP, DEHP, DINP, DIDP	[PDS] ₀ : 2.3 × 10 ^{−2} mM, PDS/PAEs: 1/1, dose: 1.67 g/L, volume: 40 mL, T: 30 °C, pH: 3	–	83% PAEs degradation; SO ₄ ^{•−} was dominant radical	[34]
	Fe@N-BC	Fishbone, urea, FeCl ₃ ·6H ₂ O, FeCl ₂ ·4H ₂ O	DEHP	DEHP/PMS: 1/1, [PMS] ₀ : 1 × 10 ^{−5} M, dose: 3.3 g/L, t: 1 h, T: 25 °C, pH: 3	–	90% removal; SO ₄ ^{•−} , •OH, O ₂ ^{•−} , and ¹ O ₂ were main ROS	[101]
Photocatalysis	CB-TiO ₂	Carbon black, TiO ₂	DnBP	[DnBP] ₀ : 15 mg/L, pH: 6.4, T: 25 °C, 15 W low-pressure UV lamp: 254 nm	>73% in 0.5 h	CB-TiO ₂ film was 1.4 times higher than that of TiO ₂ film in degradation	[104]
	MWCNTs/TiO ₂	MWCNTs, C ₁₂ H ₂₈ O ₄ Ti	DMP	[DMP] ₀ : 1 mg/L, dose: 0.5 g/L, 96 W UV lamp	–	Efficiency of MWCNTs/TiO ₂ (97%) > TiO ₂ (71%)	[30]
	TiO ₂ /C aerogel	Glucose, C ₁₂ H ₂₈ O ₄ Ti	DMP	[DMP] ₀ : 2 mg/L, dose: 0.1 g/L, 300 W Xe lamp	–	Degradation of >83%	[110]
	MWCNTs/TiO ₂	MWCNTs, C ₁₂ H ₂₈ O ₄ Ti	DMP	[DMP] ₀ : 1 mg/L, dose: 0.5 g/L, 96 W UV lamp	–	MWCNTs/TiO ₂ synthesized at 450 °C showed 97% efficiency; •OH was dominant ROS	[28]
	CuCr@BC, CuCr@CNTs	BC, CNTs, Cr(NO ₃) ₃ ·9H ₂ O, Cu(NO ₃) ₂ ·3H ₂ O	DMP	[DMP] ₀ : 15 mg/L, dose: 1.5 g/L, pH: 8, light intensity: 50 W, ultrasonic power: 150 W	–	Complete degradation (100%) in 0.5 h; •OH and O ₂ ^{•−} were prominent ROS	[99]
	GR-WO ₃ /TiO ₂	GO, Ti foil, WO ₃	DMP	DMP: 40 mL, pH: 3.13–10.09, 150 W Xe lamp	45–80% in 2–3 h	Degradation rate was 75.1%; •OH and O ₂ ^{•−} were dominant ROS	[102]
	C ₆₀ fulleranol modified Fe ³⁺	C ₆₀ fulleranol, Fe ³⁺	DEP	[DEP] ₀ : 2.22 mg/L, [PMS] ₀ : 0.5 mM, Fe ³⁺ : 40 μM, fulleranol: 10 mg/L, pH: 3.5, 420 nm light intensity 0 or 750 lx	–	58.8% degradation in 1 h increased to 100% under visible light; SO ₄ ^{•−} and •OH were active radicals	[114]
	Nanorod ZnO/SiC	Zn(CH ₃ COO) ₂ ·2H ₂ O, SiC	DEP	[DEP] ₀ : 10 mg/L dose: 0.5 g/L, 8 W UV photoreactor: 254 nm	–	>90% degradation under UV and visible light	[109]
	PANI/CNT/TiO ₂	C ₁₂ H ₂₈ O ₄ Ti, CNT, polyaniline	DEP	[DEP] ₀ : 1 mg/L, pH: 5 and 7, 10 W LED array: 410 nm and 455 nm	–	41.5–59.0% and 44.5–67.4% degradation with sol-gel hydrolysis and hydrothermal synthesis, resp.	[29]
	g-C ₃ N ₄ /Bi ₂ O ₂ CO ₃ , g-C ₃ N ₄ /BiOCl	Melamine, guanidine hydrochloride, Bi(NO ₃) ₃ ·5H ₂ O	DnBP	[DnBP] ₀ : 5 mg/L, dose: 1 g/L, Light intensity: 1 mW/cm ² , halogen tungsten lamp (500 W): 420 nm	43 and 38% on g-C ₃ N ₄ /BiOCl and g-C ₃ N ₄ /Bi ₂ O ₂ CO ₃ , resp. in 4 h	60% degradation in 5 h	[103]

(continued on next page)

Table 3 (continued)

Removal method	Met@C	Precursor	PAEs	Operation conditions	TOC mineralization	Degradation rate and key findings	Ref.
Fenton/ Fenton-like process	Nb ₂ C MXene/Fe ³⁺	Nb ₂ AlCl, AgCl, NaCl, KCl	DMP	[DEP] ₀ : 20 μM, H ₂ O ₂ : 1 mM, dose: 200 mg/L, Fe ³⁺ or Fe ²⁺ : 20 μM, pH: 3.2	37.3% in 4 h	Completely removal in 40 min; [•] OH was main ROS	[47]
	Fe/Co@NOPC	Wine lees, Co (NO ₃) ₂ ·6H ₂ O, Fe (Cl) ₂ ·6H ₂ O, KHCO ₃	DMP	[DMP] ₀ : 5 mg/L, dose: 5 mg, pH: 3, V: 0.7 V, t: 5 h	40% in 5 h	89% removal; [•] OH at cathode contributed to degradation	[40]
	Fe ₃ O ₄ @NOPC	Lees powder, KHCO ₃ , FeCl ₃ ·6H ₂ O, FeSO ₄ ·7H ₂ O	DMP	[DMP] ₀ : 10 mg/L, pH: 3, V: 0.7 V, t: 5 h	50% in 5 h	70% degradation efficiency; [•] OH was major while electro-oxidation and adsorption were minor contributors	[33]
	Co@NSC	Melamine, L-cysteine, CoCl ₂	DMP	[DMP] ₀ : 10 mg/L, H ₂ O ₂ : 3.21 mmol/L, pH: 3	33.3–38.9 % in 0.5 h	100 % degradation efficiency; [•] OH and ¹ O ₂ were main ROS	[37]
	Biphase Co@C	Co ₃ O ₄ , CaCO ₃ , LiCl, KCl	DEP	[DEP] ₀ : 5 mg/L, dose: 4 mg/L, PMS/DEP: 40/1, T: 25 °C	80% in 1 h	100% degradation efficiency; SO ₄ ^{•-} , [•] OH, O ₂ ^{•-} , and ¹ O ₂ were main ROS	[106]
	Electrolytic core-shell Co@C	CoC ₂ O ₄ ·2H ₂ O, Li–Na–K–CO ₃	DEP	[DEP] ₀ : 5 mg/L, [PMS] ₀ : 0.46 mM, dose: 50 mg/L, pH: 3.5, T: 25 °C	78.9 in 1 h	100% degradation efficiency; SO ₄ ^{•-} , [•] OH, O ₂ ^{•-} , and ¹ O ₂ co-degraded DEP	[105]
	FeCuC	Fe(III) nitrate acetylacetonate, Cu(II) acetate monohydrate	DMP	[DMP] ₀ : 50 mg/L, Na ₂ SO ₄ : 0.05 M, pH: 3	88–99% in 4 h	99% degradation efficiency; [•] OH was main ROS	[100]

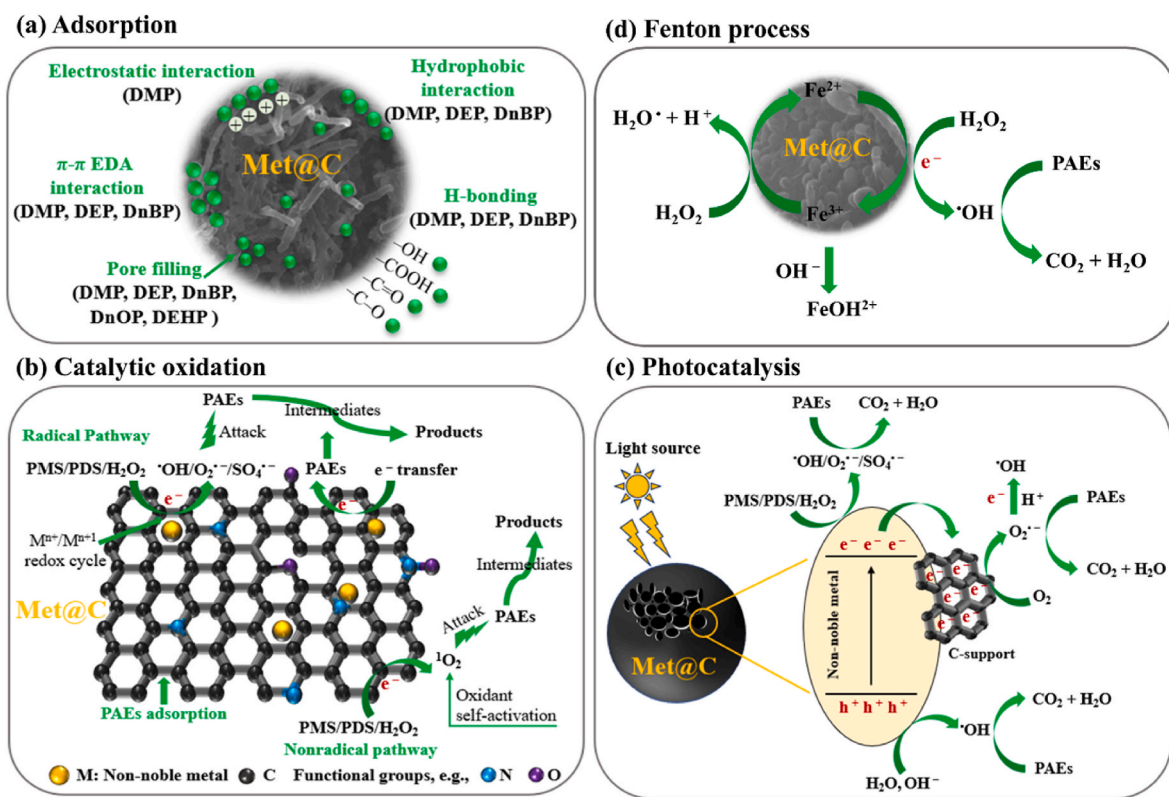


Fig. 3. Removal mechanisms of PAEs by Met@C: adsorption (a), activation oxidation (b), photocatalysis (c), and Fenton/Fenton-like process (d).

3.2. AOPs for PAEs removal

This section reviews the applications of Met@C for PAEs removal from water via activation oxidation, photocatalysis, and Fenton/Fenton-like processes, and the underlying mechanisms. Table 2 presents a comparison of these AOPs for PAEs removal.

3.2.1. Activation oxidation

Met@C is used as activator for PMS, PDS, and H₂O₂ to produce active radicals of SO₄^{•-} (redox potential: 2.5–3.1 V), [•]OH (1.8–2.7 V), and superoxide (O₂^{•-}: 2.4 V), and nonradicals (singlet oxygen (¹O₂: 0.81 V)) for PAEs oxidation via radical and nonradical pathways [107,108]. Fig. 3b shows removal mechanisms and Fig. 4b shows removal performance of PAEs by Met@C via activation oxidation. A higher redox potential suggests that radicals are more thermodynamically promising for

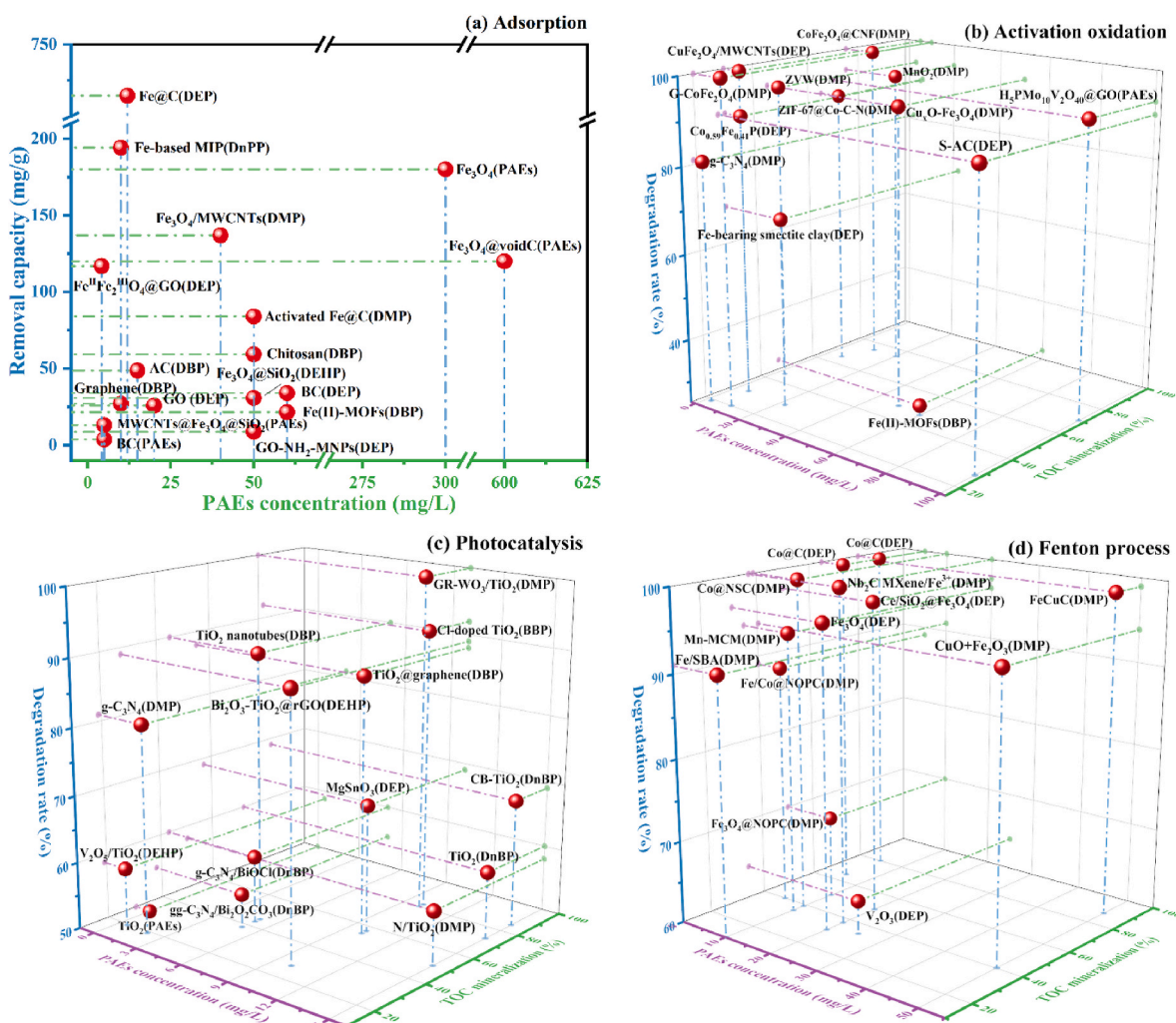


Fig. 4. Removal performance of PAEs by Met@C and other composites: adsorption (a), activation oxidation (b), photocatalysis (c), and Fenton/Fenton-like process (d).

oxidizing OCs, as it is a key thermodynamic property [10]. Enthalpy of formation for $\bullet\text{OH}$ (8.93 kcal/mol) is much higher than that for $\text{SO}_4^{\bullet-}$ (-145 kcal/mol), signifying that $\bullet\text{OH}$ is more reactive and prone to participate in multiple reactions [10]. However, PS-activation oxidation generates $\text{SO}_4^{\bullet-}$ radicals with higher oxidative potential and longer lifetime (30–40 μs) than other activation oxidation processes. $\text{O}_2^{\bullet-}$ and $^1\text{O}_2$ are less reactive but contribute to degradation of PAEs with other reactive oxygen species (ROS), direct e^- transfer, and surface-bound reactive species [6].

Xu et al. synthesized graphene-based CoFe_2O_4 (G- CoFe_2O_4) to effectively activate PMS for DMP degradation compared to CoFe_2O_4 , indicating cooperative features of graphene and CoFe_2O_4 for DMP degradation in G- CoFe_2O_4 /PMS process (Table 3) [98]. Graphene in catalyst acted as a matrix, adsorbing DMP molecules, but PMS ions desorbed them. DMP degradation occurred primarily via $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$ attack on benzene ring and oxidation of aliphatic chains. Approximately 24% of total organic carbon (TOC) was removed in 1 h, primarily due to oxidation of C in aliphatic chains. Table 3 highlights that BC as C-support for non-noble metal nanoparticles has shown satisfactory results for PMS/PDS activation to produce ROS for remediating PAEs in water, soil, and sediment [22,34,97]. Dong et al. synthesized Fe_3O_4 @BC to activate PDS for remediating PAEs contaminated sediments with >83% PAEs removal [34]. Oxygen functionalities enhanced e^- transfer, promoting $\text{SO}_4^{\bullet-}$ generation in Fe_3O_4 @BC/PDS process. LMW PAEs (e.g., DnBP) were easily removed owing to their faster kinetics due to better

solubility and accessibility to oxidants than HMW PAEs (e.g., DIDP, DINP, and DEHP) due to their limited reaction kinetics. PAEs investigated for activation oxidation by Met@C range from LMW PAEs [27,32,98] to HMW PAEs [22,34,101] (Table 3, S2). Furthermore, they synthesized Fe@N-BC to remove 90% of DEHP contaminated sediments, wherein Fe@N-BC effectively activated PMS for radical ($\text{SO}_4^{\bullet-}$, $\bullet\text{OH}$, and $\text{O}_2^{\bullet-}$) and nonradical ($^1\text{O}_2$) synergistic pathways to promote DEHP degradation [101]. Additionally, Fe-Ce@BC/PDS was utilized for degradation of PAEs, where redox capability of metal oxidation states, electrostatic and EDA interactions between PAEs and O_2 functionalities of C, and coordinated non-noble metal and C intrinsic features helped PDS activation by $\text{Fe}^{3+}/\text{Fe}^{2+}$ and $\text{Ce}^{4+}/\text{Ce}^{3+}$ redox cycles on surface of Fe-Ce@BC and degradation of PAEs [22]. Met@C and PMS/PDS can react with TOC; however, these studies overlooked TOC in sediments. In Table 3, several studies lack data on TOC mineralization, indicating that TOC was not quantified, likely due to matrix complexity or other analytical challenges. As shown in Fig. 4b, evaluating TOC mineralization with PAEs degradation rate can clarify whether PAEs are removed through adsorption, oxidation, or other mechanisms [45,98].

3.2.2. Photocatalysis

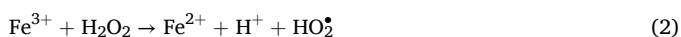
Met@C has been extensively utilized for photocatalytic degradation of PAEs with basic principle of treating PAEs under UV or visible light with or without chemicals [102,109]. Figs. 3c and 4c show photocatalytic removal mechanisms and performance of PAEs by Met@C,

respectively. MWCNTs/TiO₂ nanocomposites exhibited higher photocatalytic efficiency (97%) than TiO₂ (71%), as MWCNTs acted as e⁻ sinks and hindered recombination of e⁻ and h⁺ pairs [30]. Photo-generated h⁺ was captured by nanocomposites and transformed into •OH, boosting DMP degradation. Meenakshi and Sivasamy investigated DEP degradation by nanorod ZnO/SiC under UV and visible light irradiation, and nanocomposite exhibited >90% DEP removal [109]. As a functionalized Met@C, graphene quantum dots/Mn-N-TiO₂/g-C₃N₄ was studied for simultaneous degradation of DEP (70.4%) with photocatalytic production of H₂, and mechanism findings indicated that h⁺ was key active specie for DEP degradation [21]. Photocatalytic degradation of DMP by GR-WO₃/TiO₂ was 75.1% in 2 h, and degradation involved hydroxylation, oxidation, and dealkylation, breaking bonds and opening rings of phenolic and carboxylic acid compounds of DMP, and eventually mineralizing to CO₂ and H₂O (Fig. S2c) [102]. Mineralization rate of TOC in DnBP reached 43% on g-C₃N₄/BiOCl, whereas it was 38% on g-C₃N₄/Bi₂O₂CO₃. Compared to DnBP degradation rate (60%), it is evident that a significant portion of organic molecules were effectively mineralized during photocatalytic reactions (Fig. 4c) [103]. However, photocatalyst syntheses are relatively complex, and PAEs degradation efficiencies are relatively low, leading to a disproportionate balance between input and output performance in these photocatalytic processes.

PAEs studied for photocatalytic degradation by Met@C included DMP [28,30,102,110], DEP [21,29], and DnBP [103,104] (Fig. 4c, Table 3). High removal of LMW PAEs indicated that low molecular complexity and high solubility aided diffusion on Met@C [21,30], and moderate removal of DnBP showed slight steric hindrance due to bulkier alkyl chains and poor solubility [103]. Table S3 highlights that TiO₂ is most widely used non-noble metal oxide to synthesize Met@C for photocatalytic degradation of PAEs [111,112]. Met@C has been utilized with different light sources, including xenon (Xe) lamp [21,102,110], ultraviolet (UV) lamp [28,30,104], halogen tungsten lamp [103], and LED array [29] for satisfactory PAEs removal. Different incident light sources can cause differences in PAEs degradation. Xu et al. studied the effects of visible light (420 nm monochromatic light and LED), UV light (254 and 300 nm), UVA (320-400 nm), UVB (280-315 nm), and UVC (100-280 nm) on DMP degradation [113], wherein, C-based catalyst showed 70-90% degradation. Fe³⁺/PMS-dark process showed limited DEP removal (2.2% in 1 h, observed rate constant (k_{obs}): 0.0003 /min), but C₆₀ fullerene modified Fe³⁺/PMS-dark boosted removal to 58.8% and increased k_{obs} to 0.0142 /min – 50 times higher [114]. C₆₀ fullerene modified Fe³⁺/PMS-visible light process enhanced DEP removal from 19.5% to 100% in <1 h, raising k_{obs} from 0.0036 (in Fe³⁺/PMS-visible light process) to 0.0816 /min, demonstrating its significant role in improving degradation efficiency under dark and visible light (750 lx) conditions.

3.2.3. Fenton/Fenton-like processes

Fenton oxidation produces reactive •OH to oxidize OCs with high reaction constants, easy operation, and reaction parameters with few toxic by-products [47]. Fig. 3d shows removal mechanisms and Fig. 4d shows removal performance of PAEs by Met@C via Fenton/Fenton-like process. During Fenton (H₂O₂+Fe²⁺) reaction, H₂O₂ molecules are broken down with non-noble metal (Fe, Co, Ni, and Cu) catalysts to produce •OH in initiation reaction (Eq. (1)). However, Fenton-like process produces Fe²⁺ and HO₂• (1.7 V) (Eq. (2)) in propagation reaction, which is less effective than •OH (1.8–2.7 V) in pollutant degradation.



Met@C, such as Co@C and FeCuC, showed complete removal of LMW PAEs (DMP and DEP) with >80 TOC removal, where both radicals and nonradicals contributed to PAEs degradation, showing excellent

efficiency of Met@C in Fenton-like catalysis [100,105,106]. Hu and his team thoroughly investigated PAEs degradation by heterogeneous electro-Fenton process utilizing Fe₃O₄ loaded N and O co-doped porous C (Fe₃O₄@NOPC) [33], single-metal Co loaded N and S co-doped C (Co@NSC) [37], and Fe/Co@NOPC [40] (Table 3). Fe₃O₄@NOPC, Co@NSC, and Fe/Co@NOPC as electro-Fenton cathodes showed DMP degradation efficiencies of 70, 100, and 89%, and TOC mineralization rates of 50, 40, and 40% in 5, 0.5, and 5 h, respectively, indicating that Co@NSC was the most efficient in terms of DMP degradation and TOC mineralization (Fig. 4d, Table S4). In another study, Nb₂C MXene/-Fe³⁺/H₂O₂ completely removed DMP in 40 min, and achieved 37.3% TOC removal after 4 h, demonstrating satisfactory degradation and mineralization of DMP through •OH induced nucleophilic reactions (Fig. S2d) [47]. These findings reveal that homogeneous electro-Fenton system with these catalysts achieved rapid DMP degradation, but involved complex synthesis, poor stability, and failed to achieve synchronous TOC mineralization, where TOC removal was largely due to cathodic adsorption, requiring extended reaction times. Based on these findings, there is a need to synthesize cost-effective Met@C that ensures high catalytic activity and stability, offering a promising solution for economical and efficient PAEs pollution treatment. Only LMW PAEs – DMP and DEP – have been investigated for Fenton/Fenton-like degradation by Met@C (Table 3) [37,40,105,106], indicating the need for further research on HMW PAEs.

As voltage is significantly important in electro-Fenton degradation of PAEs, effect of voltage on DMP degradation by Fe₃O₄@NOPC was studied [33]. Degradation efficiency increased and then decreased as voltage ranged from -0.5 to -0.9 V (vs. SCE). Higher voltage (-0.7 V) accelerated e⁻ transfer, enhanced Fe³⁺ reduction to Fe²⁺ on cathode to activate, and increased H₂O₂ production, thereby improving degradation efficiency. However, beyond a certain voltage, hydrogen precipitation at cathode damaged material, reducing DMP degradation. Fe/Co@NOPC showed similar electro-Fenton degradation of DMP under different voltages; however, DMP degradation was 86% at -0.5 V, which slightly increased to 89% at -0.7 V, and then mildly decreased to 84.5% at -0.9 V [40].

3.3. Coupled treatment processes for PAEs removal

Sometimes, single treatment processes are unsatisfactory for wastewater treatment. Therefore, a combination of treatment processes that can constantly remove and degrade PAEs is required. Thus, coupled treatment processes can enhance efficiency for removing PAEs compared to single treatment [24,25]. For coupled adsorption-oxidation, adsorption immobilizes PAEs with varied CCS on Met@C, and oxidation utilizes Met@C for oxidant activation to promote PAEs degradation and mineralization via radical and nonradical pathways [107]. Functionalized Met@C, H₅PMo₁₀V₂O₄₀(44)/BC, with macropores and high porosity, showed higher adsorption capacities for PAEs, including long CCS, such as DAP and DEHP. It also degraded 80-88% of PAEs in 1.5 h via tandem hydrolysis-oxidation. DEP mineralization was confirmed by 72.5 and 64.4% reductions in chemical oxygen demand (COD) and TOC, respectively, demonstrating heterogeneity and stability of H₅PMo₁₀V₂O₄₀(44)/BC [45]. Furthermore, DMP removal via coupled adsorption and electro-Fenton oxidation showed that ≈90% DMP adsorbed on surface of DMP-imprinted activated Fe@C aerogel electrode and subsequently oxidized by surface •OH formed by electro-Fenton reaction to achieve 98% DMP removal [82]. Degradation efficiency of DMP was attributed to 4 adsorption mechanisms: electrostatic interactions, hydrophobic interactions, π-π EDA interactions, and molecular imprinting affinity between template molecules and imprinted sites (Fig. S2e) [82]. DMP was adsorbed onto activated Fe@C aerogel electrode surface through these interactions, followed by oxidation catalyzed by weak surface •OH via electro-Fenton process. Degradation products, maleic and oxalic acids, were desorbed and mineralized with free •OH in solution. Enhanced catalytic activity of

electro-Fenton process was confirmed by detecting $\cdot\text{OH}$, indicating reactivity of activated Fe@C aerogel electrode in decomposing H_2O_2 . Addition of Fe^0 enhanced graphene sheets of amorphous C and π - π interactions and shifted reaction pathway of C from $2e^-$ to $4e^-$ process. Active sites supporting selective $2e^-$ O_2 reduction to H_2O_2 for C composites would improve electrode adsorption capacity and self-cleaning ability via $\cdot\text{OH}$ generation during coupled adsorption and electro-Fenton oxidation. Met@C is not only utilized in coupled adsorption-AOPs but also in other advanced coupled treatment processes for PAEs removal. Graphene-ceramic composite containing TiO_2 and Al_2O_3 showed 99% removal efficiencies for DnBP and DEHP in coupled electrocoagulation-electrofiltration with 5 mechanisms: size exclusion, electrostatic repulsion, C adsorption, electrocoagulation, and electrofiltration [115].

4. Degradation kinetics and mechanisms of AOPs for PAEs removal

4.1. Degradation kinetics

Degradation efficiency (% removal) and pseudo-1st-order (k' , /min) and 2nd-order (k , /M s) rate constants are commonly used to assess PAEs degradation. Few studies have reported data on k' of catalysts for PAEs degradation via photocatalysis. For instance, CB- TiO_2 film demonstrated a k' value of 0.0207 /min, which is 1.4 times higher than 0.0146 /min observed for TiO_2 film during photocatalytic degradation of DnBP [104], and $\text{g-C}_3\text{N}_4/\text{Bi}_2\text{O}_3\text{CO}_3$ unveiled better photocatalytic activity with k' of 0.00249 /min, which is approximately 6.3 and 2.8 times of rate constants of pure $\text{g-C}_3\text{N}_4$ and Bi_2O_3 , respectively [103]. Furthermore, k' value from a kinetic model for DMP degradation in UV/ H_2O_2 was determined to be 0.0205-0.0907 /min [117], and in UV/PS system, k' for molar fraction of PS/DBP, ranging from 100 to 500, exhibited values varying from 0.048 to 0.144 /min [118]. Overall, there is lack of data on nonradical-dominated AOPs and role of k' in PAEs degradation.

The 2nd-order rate constant addresses this issue because it inherently reflects reaction kinetics. Numerous studies have investigated kinetics of radicals and nonradicals reacting with various PAEs, compiling k values for these radicals with PAEs, and emphasizing selectivity of different ROS [119]. Both $\cdot\text{OH}$ (1.2 - 2.8×10^9 /M s) and $\text{SO}_4^{\cdot-}$ (1.6 - 7.7×10^7 /M s) [98,120] effectively contributed to DMP degradation, whereas $\text{SO}_5^{\cdot-}$ ($<10^3$ /M s) [98] was inert, which contributed negligibly to DMP degradation. In most cases, $\cdot\text{OH}$ (3.8 - 7.6×10^8 /Ms) was swift, but $\text{SO}_4^{\cdot-}$ (4.0 - 9.1×10^5 /M s) showed relatively slow reactivity. Overall, both $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ contributed to DMP degradation [98,120]. For $\cdot\text{OH}$ radicals, k of 4.4×10^9 and $\sim 10^7$ - 10^8 /M s were evaluated for reaction of DEP with $\cdot\text{OH}$ using competition kinetics, with benzoic acid [121] and 5, 5-dimethyl-1-pyrroline N-oxide [122] as reference compounds, respectively. DnBP achieved >85% mineralization in 1 h with 25 mg/h O_3 , and TOC removal exceeded 73% in 0.5 h at 50 mg/h. UV-induced $\cdot\text{OH}$ accelerated degradation via a chain mechanism, with k of 1.3×10^8 and 1.4×10^6 /M s for formic and oxalic acids, respectively, highlighting indirect radical reactions in mineralizing intermediates [104].

For radical systems, kinetic model for DMP degradation in UV/ H_2O_2 showed k of 4×10^9 /M s for reaction between DMP and $\cdot\text{OH}$ [117], k for DBP reaction in UV/PS with $\cdot\text{OH}$ was 6.3×10^9 /M s and with $\text{SO}_4^{\cdot-}$ was 1.2×10^8 /M s using competition kinetics [118], and k values for DEP reacting with $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ were determined to be 3.7×10^9 /M s and 6.4×10^7 /M s, respectively [123]. In a nonradical system, in C_{60} fullerenol modified Fe^{3+} /PMS process under visible light, $\text{O}_2^{\cdot-}$ exhibited k of 3×10^{10} /M s, which is significantly higher than those of $\cdot\text{OH}$ (6×10^8 /M s) and $\text{SO}_4^{\cdot-}$ (8×10^5 /M s) for DEP removal [114].

In $\text{G-CoFe}_2\text{O}_4$ /PMS, TOC removal decreased within 1 h, corresponding to 2.4 carbons in DMP molecule, and aliphatic chain oxidation primarily drove TOC removal initially [98]. However, least C-containing compound contributed to 20% TOC removal, suggesting further

degradation of primary intermediates into smaller molecules [32]. During DEP oxidation by CuFe_2O_4 /MWCNTs/PDS, TOC removal was slower than DEP degradation (Table S2) [27]. After 0.5 h, DEP was fully decomposed, but TOC removal reached only 23%, indicating formation of small molecular carboxylic acids via oxidative ring opening and oxidation to CO_2 and H_2O . Cu^{2+} acted as key catalytic center to activate PDS to generate Cu^{3+} and $\text{SO}_4^{\cdot-}$, and CuFe_2O_4 /MWCNTs activated PDS via nonradical mechanisms and facilitated e^- transfer to regenerate catalytic centers (Fig. S2b).

Mineralization of DEP by $\text{H}_5\text{PMo}_{10}\text{V}_2\text{O}_{40}(44)$ /BC and O_2 showed TOC (64.4%) and COD (72.5%) removal in 2 h, which were slower than DEP degradation (80.1%), confirming oxidation as dominant mechanism over adsorption [45]. Degradation mechanisms included PAEs adsorption in macropores, Brønsted acid-catalyzed hydrolysis, redox reactions via e^- transfer, ROS generation from reduced polyoxometalates interacting with O_2 , and ROS confinement in pores. Inert compounds like DAP and DEHP were degraded through adsorption followed by oxidative hydrolysis and oxidation. Incomplete mineralization of PAEs was attributed to intermediate formation (e.g., lactic acid and ethanol), highlighting limitations of fully breaking down complex intermediates. In photocatalytic degradation of DMP by $\text{g-C}_3\text{N}_4$ /PMS, 3 intermediate groups namely hydroxy, sulfooxy, and side chain oxidized intermediates were identified with 81% DMP degradation, primarily driven by $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ radicals, accompanied by 19% decrease in TOC (Fig. 4c, Table S3) [113]. This TOC reduction was attributed to further breakdown of primary intermediates, though some polar and small molecule intermediates remained unidentified.

4.2. Reaction mechanisms

Radical and nonradical species follow distinct reaction pathways when interacting with PAEs during AOPs (Table 4). Radical pathways rely on radicals ($\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$), which exhibit high redox potentials and fast oxidation rates. These radicals degrade numerous OCs through similar mechanisms, including hydrogen atom abstraction, single e^- transfer, and radical addition formation [10,124]. However, these pathways suffer from lower selectivity due to involvement of multiple reaction substances. Although they achieve higher mineralization efficiency, they require greater chemical inputs and are more dependent on pH conditions [59].

Nonradical pathway through formation of $^1\text{O}_2$ and direct e^- transfer features a relatively mild redox potential (<1.1 V), offering improved selectivity for target PAEs due to its e^- shuttling mechanism (Table 4). It is resistant to interference from background substances and operates effectively across a broad pH range (particularly in PS-based AOPs) [6]. This pathway also minimizes self- and solvent quenching, enhances oxidant utilization efficiency, reduces need for chemical inputs, and generates fewer toxic by-products (ClO_3^- , BrO_3^- , IO_3^- , and chlorinated/bromated disinfection by-products) [6,125]. Few strategies exist for controlling chlorinated and brominated species in radical-dominated AOPs such as addition of ammonia and/or chlorine and dissolved organic matter, use of H_2O_2 as a bypass reagent, and pH adjustment [125–127].

Evidence shows that radicals react faster with PAEs than nonradical species, and radical-based AOPs can achieve high removal efficiency of OCs in a short contact time [10,131]. For example, DMP was completely degraded in 0.5 h by $\text{G-CoFe}_2\text{O}_4$ [98] and CoFe_2O_4 @CNF [32] in radical-dominated AOPs due to $\text{SO}_4^{\cdot-}$ and $\cdot\text{OH}$ radicals attack of aromatic ring and oxidation of aliphatic chain. For DEP oxidation, CuFe_2O_4 /MWCNTs activated PDS by MWCNTs through nonradical mechanisms and facilitated e^- transfer to regenerate catalytic centers for complete DEP decomposition (Fig. S2b) [27]. Few studies have shown that radicals and nonradicals synergistically promote PAEs degradation. For instance, Fe@N-BC activated PMS for radical and nonradical pathways to degrade 90% of DEHP [101]. Adjusted Met@C modification significantly affects reaction mechanism pathways in OCs

Table 4
Comparison of radical and nonradical pathways for PAEs degradation during AOPs.

Characteristics	Radical system	Nonradical system	Case studies/examples
Redox potential	High ($\text{SO}_4^{\bullet -}$: 2.5–3.1 V, $\bullet\text{OH}$: 1.8–2.7 V, $\text{O}_2^{\bullet -}$: 2.4 V)	Mild (<1.1 V) ($^1\text{O}_2$: 0.81 V)	Radical: $\text{CuFe}_2\text{O}_4/\text{MWCNTs} + \text{PDS} \rightarrow \bullet\text{OH}$, $\text{SO}_4^{\bullet -} \rightarrow$ complete DEP degradation in 0.5 h [27]; advantages: deeper mineralization and faster oxidation rate Nonradical: $\text{FeMg@NCNTs} + \text{PMS} \rightarrow ^1\text{O}_2 \rightarrow >90\%$ removal efficiency of OCs [128]; advantages: high oxidant utilization, resistant to complex water matrices, and avoid self-quenching
Selectivity	Non-selective oxidation	Selective oxidation	Radical: In DEHP degradation by UV/PMS, $\bullet\text{OH}$ preferentially attacked long side chain of benzene ring and higher e^- density led to cleavage of ester groups [129] Nonradical: In DMP degradation by Co@NSC , $\bullet\text{OH}$ (and $^1\text{O}_2$) attacked benzene ring [37]; overall selective oxidation of OCs [6]
pH	pH dependent	Less pH-sensitive	Radical: Optimal PS activation at acidic pH (pH 2.1–4.9) [114]; 83% degradation of DnBP, DEHP, DINP, DIDP by $\text{Fe}_3\text{O}_4/\text{BC}$ at pH 3 [34]; optimal Fenton process at pH ~3–4 [100,105]; but performance drops at high pH in different studies Nonradical: Efficient over wide pH range [6]; >80% catalytic efficiency for wide pH range [130]
Chemical input	High	Low	Radical: High oxidant requirement (>1 mM), such as 2 mM $[\text{PMS}]_0$ for complete DMP degradation [98]; relatively higher Fe@N-BC dose (3.3 g/L) for 90% DEHP removal [101] Nonradical: Low oxidant requirement (<0.5 mM), such as 0.46 mM $[\text{PMS}]_0$ for 100% DEP degradation [105]
By-product toxicity	Relatively high toxicity	Low toxicity	Radical: Cl^- or Br^- presence in AOPs form harmful by-products (ClO_3^- and BrO_3^-), with BrO_3^- being a carcinogen [125]

Characteristics	Radical system	Nonradical system	Case studies/examples
Catalyst stability	Relatively low	Relatively high	Nonradical: Minimal toxic by-products observed in PS-based AOPs [6] Radical: Low catalyst stability due to catalyst leaching and deactivation; >74% DMP degradation by $\text{GR-WO}_3/\text{TiO}_2$ after multiple uses [102]; but high Fe leaching observed in Fe-based systems Nonradical: High catalyst stability due to absence of radicals; degradation efficiency of Fe@C/PMS decreased from 91 to 77% after multiple uses [130]
Metal Leaching	Maximal in homogeneous systems	Minimal in heterogeneous systems	Radical: Metal ions leaching from $\text{CuFe}_2\text{O}_4/\text{MWCNTs}$ was <0.4% [27], but metal-based catalysts leach in homogeneous systems (e.g., Fenton process) Nonradical: Metal leaching from Fe@C reduced in alkaline conditions [130]
LMW PAEs	High removal; better attack on aromatic ring or ester group	Moderate removal; resilience in complex matrices	Radical: 97% DMP photodegradation by $\text{MWCNTs}/\text{TiO}_2$ in 3 h UV irradiation [28] Nonradical: 100% DEP degradation by Co@C with radicals and nonradicals [105]
HMW PAEs	Moderate removal; steric hindrance slows reaction kinetics	Moderate-high removal; less scavenging by matrix species	Radical: 83% HMW PAEs degradation [34]; but sometimes partial (and slow) PAEs degradation due to bulky structures Nonradical: 90% DEHP removal with radicals and nonradicals via surface-bound active sites [101]

degradation. While Fe@NCNTs/PMS followed radical pathway involving ROS, FeMg@NCNTs/PMS shifted the mechanism to non-radical pathway dominated by $^1\text{O}_2$, indicating that Mg doping entirely transitioned activation pathway of PMS [128]. Redox potential of radical/nonradical pathways can be tuned by adjusting Met@C structure and composition (i.e., heteroatom doping and defects) as well as its physicochemical properties (i.e., conductivity and zeta potential) [6]. Non-noble metals and oxides are widely used as target catalysts for PS activation in homogeneous and heterogeneous systems. PS is activated into ROS through cleavage of peroxide O–O bonds, a process similar to H_2O_2 activation in Fenton/Fenton-like reaction [6]. Additionally, PS can generate nonradicals, formation of these reactive species is influenced by physicochemical properties of Met@C , including crystalline structure, defective sites, electrical conductivity, and electronic configuration [12,132].

Mineralization rates of OCs, measured by residual TOC, are typically

higher with radicals than with nonradicals ($^1\text{O}_2$) [131]. While some studies used TOC reduction as an indicator of $^1\text{O}_2$ generation [133], but others argued that $^1\text{O}_2$ lacks strength to mineralize many OCs [134]. Thus, TOC removal rate might not be a reliable indicator for differentiating between radical and nonradical pathways, as adsorption of intermediates on Met@C surface can also lead to reduction in dissolved organic C [6]. Functionalized but complex Met@C such as G-CoFe₂O₄ [98], CuFe₂O₄/MWCNTs [27], H₅PMo₁₀V₂O₄₀(44)/BC [45], Fe₃O₄@-NOPC [33], Co@NSC [37], Fe/Co@NOPC [40], g-C₃N₄/Bi₂O₂CO₃ [103], and Nb₂C MXene/Fe³⁺ [47] demonstrated enhanced PAEs removal in a short time but these Met@C exhibited low TOC removal over extended time, indicating incomplete and potentially misinterpreted information about reactants, intermediates, and products (and toxic by-products) involved in PAEs removal processes (Fig. 4b, 4c, 4d).

Improper use of chemical, quenching, and scavenging agents in electron paramagnetic resonance (EPR) spectroscopy and chemical quenching can mislead radicals, nonradicals, and e^- transfer processes in AOPs [10]. This can result in incorrect conclusions regarding ROS regeneration and PAEs degradation pathways identification. On one hand, there is a need to simplify Met@C synthesis using easy and straightforward methods with minimal chemical usage. On other hand, identifying radicals, nonradicals and e^- transfer processes in AOPs requires advanced techniques such as linear sweep voltammetry (LSV), quantitative structure-activity relationships (QSARs), DFT and MD simulations, and open circuit potential measurements [6,129]. These techniques are essential to clarify mechanisms of PAEs removal and degradation by Met@C.

4.3. Computational insights into PAEs removal mechanisms: role of DFT and MD simulations

In recent decades, DFT and MD simulations have provided atomic-level understanding of thermodynamic behaviors and electronic

structures governing PAEs-Met@C interactions, thereby guiding rational design of efficient Met@C [37,38].

4.3.1. DFT-based mechanistic insights

DFT calculations are essential tools for synthesis of Met@C and design of removal processes, as they offer valuable insights into electronic structures, bond strengths, and reaction mechanisms at atomic-level [36,38]. To elucidate degradation mechanisms of OCs by Met@C in AOPs, DFT calculations demonstrated that in Fe@C/PMS, OCs oxidation was primarily driven by surface-bound radicals ($\text{SO}_4^{\cdot-}$ and $\cdot\text{OH}$), $^1\text{O}_2$, and direct e^- transfer [130], and FeMg@NCNTs/PMS transitioned to nonradical pathway (dominated by $^1\text{O}_2$) from radical pathway observed in Fe@NCNTs/PMS [128], showing suitability of Met@C for OCs removal.

DFT calculations, including electrostatic potential (ESP) maps, adsorption energy analysis, Fukui functions, and electronic properties, frontier orbitals, and geometric structures of Met@C, are crucial for rational design of Met@C with enhanced reactivity and selectivity toward specific PAEs congeners. ESP maps reveal electrostatic interactions between PAEs and Met@C, guiding adsorption propensity [37,135]. DFT calculations with LC-MS explored DMP degradation pathway via electro-Fenton by Co@NSC, wherein ESP map shows positive (blue) region on benzene ring and negative (red) region on branched chain, indicating that benzene ring, where highest occupied molecular orbital (HOMO) exists, is more susceptible to electrophilic free radical ($\cdot\text{OH}$) attacks and branched chain, where lowest unoccupied molecular orbital (LUMO) exists, is more susceptible to nucleophilic attacks (Fig. 5a) [37,136]. Results of frontier orbitals were consistent with ESP, indicating that $\cdot\text{OH}$ was main ROS acting as electrophilic substance to attack benzene ring during reaction [37].

Adsorption energy analysis quantifies binding strength and helps compare efficacy of different Met@C systems, offering a quantitative measure of PAEs-Met@C compatibility during AOPs [122]. In multi-stage adsorption, DFT calculations showed that binding energies

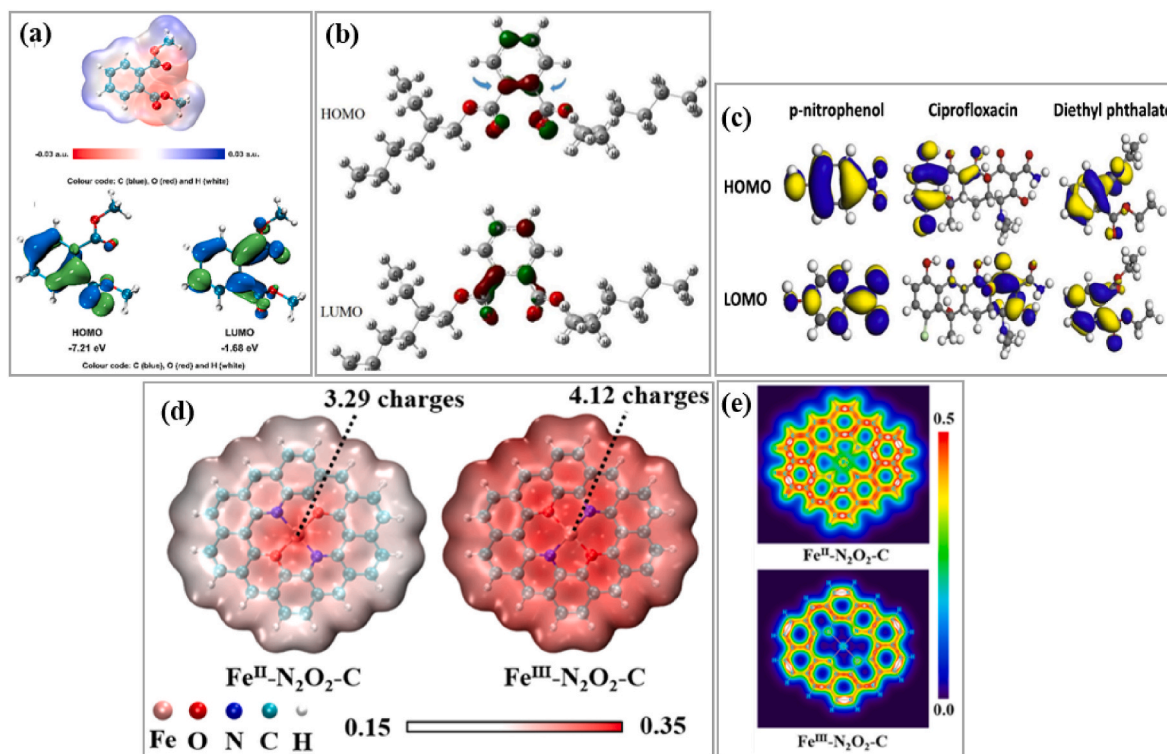


Fig. 5. ESP mapping and HOMO and LUMO orbitals of DMP (a) [37], frontier electron orbitals (HOMO and LUMO) of DEHP as calculated by Gaussian 09 W program (b) [129], LUMO and HOMO orbitals of 4-NP, CIP, and DEP (c) [21], ESP maps of Fe^{II}-N₂O₂-C and Fe^{III}-N₂O₂-C (d) and e^- density 0.64 Å above Fe^{II}-N₂O₂-C and Fe^{III}-N₂O₂-C plane (e) [135].

of DEP and DBP dimers were $-45.59 \text{ kJ mol}^{-1}$ and $-59.71 \text{ kJ mol}^{-1}$, respectively, and MD simulations indicated that π - π EDA interactions were dominated removal mechanism for both PAEs [137]. Combined with material characterization, DFT calculations revealed that N-doped sites in N@C-based catalysts enhanced catalytic activity due to optimized bond lengths, charge distribution, and adsorption energy [138].

Fukui functions identify specific reactive sites on PAEs and Met@C surfaces, wherein f^- , f^+ , and f^0 represent susceptibility of an atom toward electrophilic, nucleophilic, and free radical attack, respectively [47,139]. Previous studies based on Fukui indices have shown that benzene ring is susceptible to electrophilic and free radical attacks, whereas branched chain is more prone to nucleophilic attack [37]. For degradation of DMP by $\text{Nb}_2\text{C MXene/Fe}^{3+}/\text{H}_2\text{O}_2$, Fukui functions and Hirschfeld charges for DMP and its intermediates were considered to identify key states of N, N+1, and N-1 for each element for electrophilic and nucleophilic attacks [47]. DMP degradation was primarily initiated by $\bullet\text{OH}$, leading to nucleophilic reactions that progressively produce smaller molecules. For f^+ of DMP, most nucleophilic attack sites were O (4), C (9), C (10), and C (12) (Fig. S2d). Furthermore, DFT calculations have been used to calculate frontier orbitals and Fukui indices to examine e^- transfer in AOPs of PAEs [106,135,140].

Electronic properties, frontier orbitals, and geometric structures of Met@C obtained via DFT calculations provide valuable qualitative and quantitative insights, including band structure, charge and spin density, and energy band [141]. HOMO and LUMO are examined for PAEs removal, as they correspond to molecule's e^- donating and e^- accepting abilities, respectively [37,128]. Generally, sites with large $2\text{FED}_{\text{HOMO}}^2$ are more likely to undergo direct oxidation by e^- extraction, while $\bullet\text{OH}$ addition favours sites with high $\text{FED}_{\text{HOMO}}^2 + \text{FED}_{\text{LUMO}}^2$ values [129,142]. For UV/PMS degradation of DEHP, C (1) exhibited highest $\text{FED}_{\text{HOMO}}^2 + \text{FED}_{\text{LUMO}}^2$ value, indicating that $\bullet\text{OH}$ preferentially attacks this position, leading to subsequent cleavage of long side chain of benzene ring (Fig. 5b) [129]. However, highest e^- density was localized on several O atoms (such as O (8), O (9), O (19), and O (20)), suggesting that these sites are most susceptible to oxidation, leading to cleavage of ester groups. In Co@C/PMS system, phenyl ring of DEP exhibited substantial contributions to both HOMO (-1.78 eV) and LUMO (-5.52 eV) energy levels, indicating that phenyl ring served as primary reactive site for $\bullet\text{OH}$ and $\text{SO}_4^{\bullet-}$ during chemical reactions with DEP [106]. Similarly, HOMO and LUMO of p-nitrophenol (4-NP), ciprofloxacin (CIP), and DEP explored their photodegradation rates by graphene quantum dots/Mn-N-TiO₂/g-C₃N₄ (Fig. 5c) [21]. Higher degradation rate of 4-NP compared to CIP and DEP was attributed to its smaller molecular size. In DEP, overlapping e^- clouds in LUMO and HOMO promoted recombination of e^- and h^+ , whereas in CIP, partial separation of e^- clouds resulted in better e^- and h^+ separation, leading to higher CIP degradation than DEP [21].

Theoretical evaluation of Met@C structures has also been performed using DFT calculations [105,106]. Theoretical calculations confirm that Co nanoparticles encapsulated with graphitized C shell derived from CO₂ reduced work function, elongated O-O bond length of PMS, and facilitated charge transfer from Co@C to PMS. These findings aligned with catalytic experiments showing that Co@C was significantly more reactive than bare Co for PMS activation and DEP degradation [105]. To investigate geometry of functionalized C-based materials, Liu et al. identified several highly reactive active sites by combining N bond length change and adsorption energy: C (42) of single-vacancy defect graphene (GR1), C (7) of bipyridine N-doped GR1, C (14) of bipyrrrole N-doped GR1, C (41) of mono-graphite N-doped GR1, C (40) of binary S-doped GR1, C (35) of temporary double-vacancy defect graphene (GR2), C (34) of bi-graphite N-doped GR2, and C (7) of binary S-doped GR2 (Fig. S3a-h) [143]. Subsequently, intermediates were adsorbed at these active sites to investigate their effects on catalytic reaction.

4.3.2. MD-based mechanistic insights

Although DFT provides static state atomic-level insights into

idealized structures, MD simulations offer a time-resolved perspective on dynamic states of molecular motion and interfacial behavior under realistic conditions. By incorporating thermal fluctuations and solvation effects, MD simulations validate DFT-derived mechanisms and ensure their dynamic plausibility [135]. MD is a systematic method used to study relationships among molecular functions, particle motion, and structure [144,145]. Recent studies have used MD to i) investigate diffusion dynamics of PAEs near Met@C surfaces, revealing how mobility affects adsorption kinetics, ii) monitor evolution of intermolecular interactions over time, including hydrogen bonding, π - π stacking, and hydrophobic interactions, and iii) simulate adsorption-desorption equilibria under variable environmental conditions (e.g., pH, temperature, and ionic strength), mimicking real-world water matrices [137,146,147].

Our group investigated intermolecular interactions of PAEs in multi-stage adsorption using experimental studies and computational simulations [137]. DFT and MD simulations revealed that DBP formed stable aqueous clusters at low concentrations due to its lower binding energy and stronger interactions than DEP. These clusters adsorbed rapidly onto BC as units, then broke up and re-equilibrated. To elucidate the action of O₃ and Fe-N₂O₂-C, DFT calculations were used to predict active sites based on electrostatic effects and e^- density, revealing Fe^{III}-N₂O₂-C as more reactive than Fe^{II}-N₂O₂-C due to higher positive ESP and lower e^- density around Fe atom (Fig. 5d and 5e) [135]. Central Fe in Fe^{III}-N₂O₂-C exhibited highest positive ESP, enhancing its nucleophilicity and attraction to nucleophilic end atom of O₃, making it more effective active site. MD simulations confirmed this prediction and observed O₃ decomposition during process.

5. Metal leaching of Met@C for PAEs removal

Metal-based catalysts (metal oxides such as Co₃O₄, MnO₂, and Fe₃O₄) exhibit high catalytic efficiency in activating PMS/PDS at $\mu\text{g/L}$ concentrations to degrade OCs [32,148]. However, their nanosize complicates post-reaction separation, risking environmental contamination, and poor stability and metal ion leaching further limits practical viability [122]. Correlation between structure and property suggests that encapsulating metal nanoparticles with C layers and supports preserves their intrinsic catalytic activity, while simultaneously delaying metal leaching and Met@C aggregation and improving their stability [32,101,149]. Compared to Met@C [27,33], non-noble metal oxide composites exhibited higher metal leaching, with levels ranging from 1.25 to 2.30 mg/L [150].

Stability of Met@C correlates with metal leaching levels, with Fe demonstrating superior stability due to minimal leaching [151]. Fe@C immobilizes on diverse supports – C aerogels [82], BC [34], biomass porous C [33], MWCNTs [27], and graphene [39] functionalized with iron – to form composites that minimize Fe leaching ($<0.1 \text{ mg/L}$). For instance, Fe₃O₄@NOPC exhibited high stability with minimal Fe leaching ($<0.1 \text{ mg/L}$) under acidic conditions, effectively degrading DEP via electro-Fenton process [33]. CuFe₂O₄/MWCNTs demonstrated negligible leaching of metal ions ($<0.4\%$) during reuse, highlighting their oxidation resistance in DEP degradation [27]. Fe@C shows negligible Fe leaching ($\leq 2.08\%$ at pH 2.6) across pH 4-11.5, attributed to protective C layer shielding Fe²⁺ particles [61]. Another study confirmed that leached Fe concentrations remain below 0.03 mg/L at pH 3-9 [152]. Besides, leaching from Cu@C [153] and Cu@S [122] showed negligible release ($<1\%$ and $<0.01 \text{ mg/L}$ Cu²⁺ released, respectively), with degraded filtrate showing no toxicity to cells and organisms. Similarly, Zn, often bound as Zn-O or Zn-N complexes, showed no significant leaching during AOPs [154]. Among non-noble metals, Co-based catalysts exhibit highest catalytic efficiency but suffer from leaching, which is a critical drawback. To address this, hybridizing Co with other non-noble metals (CoFe₂O₄, which forms strong Fe-Co bonds [155]) and incorporating supports (C nanofibers [32] and N-doped C [156,157]) can effectively suppress Co leaching during AOPs.

Additionally, leaching of Co decreases with increasing pH from acidic to neutral conditions in AOPs [158].

Leaching of metal ions in heterogeneous catalysts for water treatment poses significant concerns, as reducing it mitigates secondary pollution risks [156]. Previous studies have reported relatively higher amounts of metal residues during removal of contaminants by metallic catalysts supported on AC (18.1%) [159] and CNTs (15.9%) [160]. Aggregation, non-uniform dispersion of metal particles, and metal leaching remain persistent challenges that require resolution. Functional groups on C-based materials effectively anchor metal ions (Fe^{2+} , Co^{2+} , Cu^{2+} , and Ni^{2+}), inhibiting their leaching while stabilizing them [32]. Met@C composites, such as $\text{H}_5\text{PMo}_{10}\text{V}_2\text{O}_{40}(44)/\text{BC}$ (no HPMoV leaching) [45] and $\text{Nb}_2\text{C MXene}/\text{Fe}^{3+}$ ($<30 \mu\text{g/L}$ of Nb leaching) [47], are considered green functionalized materials because they showed no or least metal leaching during PAEs removal. Overall, the concentration of leached metal ions remained below the China National Water Quality standards (0.1 mg/L) under tested conditions [33,47,152]. Leaching of metals from Met@C during OCs degradation often occurs, but limited studies have reported precise metal ion dissolution in AOPs [156,161]. Further research is needed to evaluate potential long-term leaching of metal ions to better understand catalytic activity, stability, and PAEs removal capacity of Met@C in complex environments at large-scale.

6. Real-world applications of Met@C for PAEs removal

Transition of Met@C for PAEs removal from lab to real-world applications is required. In real wastewater, numerous complex background matrix components, such as heavy metals, organic impurities, multiple ions, and pH variation, along with target PAEs can influence performance (see Text S2 of Supplementary Information). Such components can destroy C matrix, thereby affecting catalytic activity and stability of Met@C. Maintaining performance of Met@C in practical applications requires careful selection of suitable metal and C precursors prior to synthesis. To check feasibility of $\text{MWCNTs}@/\text{Fe}_3\text{O}_4/\text{SiO}_2$ for PAEs removal, 5 mg/L of DMP, DEP, and DnBP were spiked into groundwater, surface water, and domestic sewage [95]. Removal efficiencies of DMP, DEP, and DnBP in groundwater, surface water, and domestic sewage were 60.0, 74.6, and 94.1%; 66.3, 84.0, and 96.9%; and 51.4, 75.4, and 94.7%, respectively. DMP showed lowest removal efficiency, particularly in domestic sewage (51.4%), suggesting that molecular size and CCS of PAEs, which increased from DMP to DnBP, may affect their removal, with surface water showing highest PAEs removal. Zhuang et al. applied $\text{Fe}_3\text{O}_4/\text{MWCNTs}$ for DMP adsorption from municipal wastewater [31]. For a DMP concentration of 40 mg/L , influent from secondary treatment showed TOC and COD removal efficiencies of 17.95 and 22.60%, while effluent showed 18.76 and 30.51%, respectively, after adsorption. With the same DMP concentration, deionized water, influent, effluent, and tap water showed removal efficiencies of 42.25, 17.11, 19.19, and 40.54%, respectively, indicating reduced efficiency with more complex water samples.

A study used functionalized $\text{GO-NH}_2\text{-MNPs}$ to remove PAEs (DMP, DEP, DnBP, and BBP) and other compounds (estriol (as model estrogens) and fluorene (as model PAHs)) from water sources, including tap water, mineral water, ice block water, and Pearl River water (Southern China) [80]. With spiked water containing 0.02 mg/L of DMP, DEP, and estriol (except Pearl River water), and 0.04 mg/L of DnBP, BBP, and fluorene, $\text{GO-NH}_2\text{-MNPs}$ achieved $>99\%$ removal efficiency of analytes (PAEs + estriol + fluorene) in all water samples. Adsorption was attributed to $\pi\text{-}\pi$ interactions between benzenoid-like structures of analytes and $\text{GO-NH}_2\text{-MNPs}$, suggesting their effectiveness in removing PAEs (benzenoid-like compounds) from environmental waters. For coupled treatment process, $\text{H}_5\text{PMo}_{10}\text{V}_2\text{O}_{40}(44)/\text{BC}$ removed DEP ($8.6 \mu\text{g/L}$) and DAP ($4.5 \mu\text{g/L}$) from secondary effluent, achieving removal efficiencies of 62.2 and 78.1%, respectively, in 1 h. A couple of PAEs did not impact BC performance, attributing to persistence of doped structure for strong adsorption and oxidation capabilities in aerobic oxidation of PAEs [45].

In the static test, adsorption efficiency of chitosan for PAEs followed the order diheptyl phthalate (DHPP) $>$ DBP $>$ DEHP $>$ DPP $>$ DEP $>$ DMP [162]. Effects of temperature, pH, Ca^{2+} ions, and salinity on plastic manufacturing wastewater (Tainan, Taiwan) were investigated in the dynamic test. Temperature optimization showed maximum adsorption at 40°C and minimal desorption differences between 25°C and 40°C . pH (6–9) revealed optimal performance at pH 8–9, leading to pH 8 recommendation for real wastewater. Ca^{2+} ions influence was significant only at concentrations $\geq 400 \text{ mg/L}$, while salinity showed no measurable impact on adsorption efficiency [162]. Surface modification of AC (55 mg/g) with Cu (117 mg/g) or tetrabutylammonium (92 mg/g) significantly enhanced its adsorption capacity, facilitating 98% PAEs removal from industrial wastewater [163]. However, such modifications considerably increased cost of modified adsorbents, indicating that life-cycle cost analysis of Met@C must be conducted. Cost evaluation of PAEs removal by Met@C is also described in Text S5 of Supplementary Information.

While numerous lab-scale studies have reported high PAEs removal efficiencies, few pilot-scale and field trials have documented satisfactory removal ($>70\%$) of multiple PAEs [164–166]. Integration of PAEs removal by Met@C with existing wastewater treatment infrastructure (e.g., as a post-treatment step) is a promising avenue. Lower ecotoxicity compared to noble metals and ease of post-treatment integration into existing water treatment infrastructures, such as adsorption systems, biological treatments, and hybrid AOPs can enhance scalability and cost-effectiveness of mass production of Met@C and their use for large-scale remediation of PAEs contaminated water systems.

Influencing factors, regeneration, ecotoxicity assessment, and cost evaluation of PAEs removal by Met@C are described in Text S2, S3, S4, and S5, respectively, of Supplementary Information. Furthermore, Figs. S1, S2, S3, and S4 and Tables S1, S2, S3, S4, S5 and S6 are described in Supplementary Information.

7. Perspectives, future challenges, and opportunities for PAEs removal by Met@C

Despite extensive research on PAEs removal and exceptional performance of Met@C, substantial research challenges concerning its synthesis, removal design, and environmental applications persist.

- i) **Suitability of Met@C for PAEs removal:** It is necessary to determine which Met@C is more suitable for PAEs removal. PAEs are comparatively e^- donating compounds due to resonance and inductive effects of ester groups and alkyl chains. Met@C should possess properties that enable generation of strong electrophilic species or facilitate oxidative processes to target and breakdown PAEs. Metal dopants or functional groups on C surface can exhibit e^- withdrawing properties to complement e^- donating nature of PAEs. For instance, doping with metals such as Zn or Ni can introduce e^- deficient sites that attract e^- rich PAEs.
- ii) **Unique reactivity of Met@C for PAEs degradation:** PAEs immobilization in water systems is a considerable problem; adsorption of PAEs onto Met@C and subsequent degradation is necessary. Graphitic structures and metal species along with surface functional groups as additional active sites for adsorption and catalysis (especially hydrophilic functional groups, e.g., $-\text{COOH}$, $-\text{OH}$, or $-\text{NH}_2$, to improve water wettability and facilitate interactions) would have unique reactivity for PAEs with varying CCS that can guide efficient PAEs removal and Met@C design.
- iii) **Stability of Met@C in real-world applications:** Several high-performance composites have recently failed because of their low stability. Non-noble metal composites can oxidize and form aggregates, which affect their removal performance. To ensure chemically and structurally stable Met@C in complex environments, long-term metal dissolution and oxidation of Met@C must

be addressed and metals like Fe or Co doped into robust C matrices (e.g., graphene, CNTs, or BC) can maintain their activity for longer storage and multiple uses for PAEs removal.

- iv) **Adjustment of coordinated environment:** Coordination number and environment of non-noble metal active centers are crucial for determining electronic properties and catalytic performance. It is essential to develop strategies to precisely control coordination geometry and isolated atoms in reaction center by introducing vacancies and structural defects into C matrix.
- v) **Role of primary and secondary radicals in PAEs degradation:** Few studies have investigated reactivity of Met@C for removal of PAEs with varied CCS and role of (primary and) secondary radicals in PAEs degradation during AOPs [119]. However, published results are inconsistent due to complex interactions among Met@C, oxidants, and PAEs, highlighting need for further exploration of primary and secondary radicals in PAEs degradation.
- vi) **Theoretical insights into PAEs removal mechanisms:** DFT calculations help to characterize Met@C properties and PAEs removal for related molecular configurations, charge distributions, and adsorption energies [135]. Key factors such as energy changes, reaction kinetics, and e^- transfer in process remain unclear and dynamic interactions between PAEs and Met@C are rarely demonstrated, highlighting need for guiding design and reducing costs. As computational tools evolve, they will be crucial in advancing analytical chemistry approaches to environmental remediation, particularly in designing and validating next-generation Met@C for emerging contaminants, such as PAEs. Combining DFT with methods such as *in-situ* infrared spectroscopy, enhanced sampling MD techniques, EPR, and other computational techniques (independent gradient and Gibbs free energy) and integrating machine learning with DFT/MD data can validate results and accelerate understanding of design of PAEs removal by Met@C.

8. Conclusions

This review highlights growing environmental concerns posed by PAEs and challenges associated with their removal from contaminated water systems. Adsorption, AOPs, and coupled treatment processes are usually considered better strategies for remediating PAEs contaminated water, each governed by distinct mechanisms. Although Met@C composites offer enhanced PAEs adsorption capacities and specific benefits (high surface area and selectivity), they also face challenges, such as synthesis complexities, potential environmental risks of used Met@C, regeneration, and structural stability issues that need to be addressed for progressive PAEs removal. Despite showing better PAEs degradation by Met@C, AOPs have limitations such as incomplete PAEs mineralization and unwanted radicals and toxic by-products formation. Coupling AOPs with other conventional and advanced processes is essential to improve breakdown of PAEs and reduce environmental risks. Moreover, theoretical modeling (such as DFT and MD simulations) complementing experimental studies can contribute to optimizing Met@C design, selecting optimal reaction parameters, reducing operation time and cost for PAEs removal by Met@C, and exploring degradation kinetics and mechanisms of AOPs for PAEs removal.

To solve technique problems in remediation of PAEs contaminated water systems, among next-generation materials, Met@C composites show promising potential for efficient PAEs removal, along with excellent stability, low metal ions leaching, and regenerative characteristics, making them suitable for real-world applications. This review provides a basis for effectively framing regulations to limit PAEs levels in water, offers insights into advanced processes recently assessed for PAEs removal and degradation, and highlights future research focus on long-term performance, life-cycle cost analysis, and large-scale validation to bridge the gap between lab findings and practical implementation of

PAEs removal by Met@C.

CRedit authorship contribution statement

Shakeel Ahmad: Writing – original draft, Validation, Software, Formal analysis, Data curation, Conceptualization. **Muhammad Faheem:** Writing – review & editing, Software, Methodology, Data curation, Conceptualization. **Rongli Yang:** Validation, Software, Methodology, Investigation. **Muhammad Waqas:** Visualization, Validation, Software. **Chenghong Ao:** Investigation, Data curation. **Junhui Zhang:** Investigation, Data curation. **Gang Chu:** Visualization, Investigation. **Muhammad Azher Hassan:** Software, Formal analysis. **Quan Chen:** Supervision, Investigation. **Bo Pan:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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